

DroidLink LED Strip Configuration Walkthrough

(DroidLink_Slave + DL_AP)

This guide explains LED setup from zero, for users who do not already know how sequence builders work.

1) What you are configuring

In DroidLink WebUI, the LED system has 4 separate parts:

1. LED strip setup (hardware details)
2. LED sections (optional named regions on the strip)
3. LED sequences (saved lighting timelines)
4. Startup/indicator behavior

Most problems happen when users mix these parts together. Keep them separate while learning.

2) Core concept before clicking anything

The strip is one long line of pixels.

- Pixel numbering starts at 0.
- If you have 30 LEDs, valid positions are 0 through 29.

A section is just a named range of those pixel numbers.

Example:

- Section A = 0 to 9
- Section B = 10 to 19
- Section C = 20 to 29

Sections cannot overlap, and they must stay inside total LED count.

3) Hardware wiring basics (must be correct first)

From WebUI guidance for this project:

- Data line goes to GPIO4.
- ESP32 ground and LED power-supply ground must be tied together.
- LED strip power comes from proper external power, not from the ESP32.

If power and ground are wrong, software changes will not fix behavior.

4) Open the correct WebUI area

1. Enter Web Config mode for the device.
2. Open `http://192.168.4.1`.
3. Open the LED Config tab.

Inside LED Config, work top-to-bottom:

1. LED strip setup
2. Test saved LED setup
3. Startup lighting
4. Web Config indicator

5) Step-by-step: LED strip setup

In 1. LED strip setup:

1. Confirm LED data connection is shown as GPIO4.
2. Set Total LEDs connected to full strip length.
3. Choose Type of LED strip (WS2812B / NeoPixel is most common).
4. Set Maximum strip brightness (global cap, 0-255).
5. (Optional) Add named sections.
6. Click Save LED setup.

Important:

- If total LED count is wrong, every section and sequence will be offset or clipped.
- Overall brightness limits the entire strip output.

6) Step-by-step: Add and save LED sections

Use sections when different physical areas need separate lighting.

1. Click Add LED section.
2. Give it a clear name (example: COIN_SLOT, FRONT_LOGIC, REAR_RING).
3. Set start/end range so it stays in bounds.
4. Repeat for more sections (up to 3 user-configured sections in this UI).
5. Make sure no ranges overlap.
6. Click Save LED setup.

Best practice:

- Build sections to match real physical zones on the droid.
- Keep names short and descriptive so sequence building is easy later.

7) Step-by-step: Test saved LED setup

Go to 2. Test saved LED setup.

1. Select target (Entire strip or saved section).

2. Pick a color preset or custom color.
3. For RGBW strips, set extra white level if needed.
4. Click Show selected color.
5. Confirm only expected LEDs respond.
6. Use Turn this section off or Turn entire strip off when done.

If wrong area lights up:

- Recheck section boundaries.
- Recheck total LED count.
- Save again, then retest.

8) LED sequence builder: what is happening

A sequence is a timeline of lighting actions.

Each action means:

- At time X ms, do action Y on target Z.

One action is not a complete show by itself. If you want change-over-time behavior, add multiple timed actions.

Example:

- 0 ms: FRONT_LOGIC -> Start scanner
- 2000 ms: FRONT_LOGIC -> Solid blue
- 5000 ms: FRONT_LOGIC -> Off

9) Step-by-step: Create LED-only sequences

Open LED Seq.

1. Click Add Action.
2. For each action set:
 - Start time (ms)
 - LED section or Entire strip
 - Action type (animation, solid color, off)
 - Effect settings (speed, brightness, color)
3. Add all actions needed.
4. Click Preview sequence.
5. Enter Sequence name.
6. (Optional) assign available command shortcut.
7. Click Save sequence and command.

Then use Run or edit saved LED sequences to run, edit, export, import, or delete.

10) Startup lighting vs normal sequences

Startup lighting is separate from normal run commands.

In 3. Startup lighting:

1. Choose saved LED sequence (or keep LEDs off).
2. Click Save startup choice.

Behavior notes:

- Command 60 restores saved startup lighting.
- Command 61 stops effects and turns LEDs off.
- Commands 62-91 are LED-only saved sequences.

11) Web Config indicator (optional)

In 4. Web Config indicator:

1. Enable indicator checkbox.
2. Select target section.
3. Select color.
4. Set duration.
5. Save indicator settings.

This is only a short "Web Config ready" light and does not consume a sequence slot.

12) Troubleshooting checklist

If LEDs do not behave as expected, check in this order:

1. Data pin really connected to GPIO4
2. Common ground between ESP32 and LED power supply
3. Correct strip type selected
4. Correct total LED count saved
5. Section ranges are valid and non-overlapping
6. Brightness not set too low
7. Testing saved setup before sequence editing

13) Teaching script for beginners

Use this short script with builders:

- "First we define what strip exists (count, type, brightness)."
- "Then we divide it into named zones (sections)."
- "Then we tell those zones what to do over time (sequences)."
- "Every sequence action is one timed instruction."
- "If we do not add an action, it will not happen."

14) One-sentence summary

Configure strip first, verify sections second, then build timed lighting actions in sequences - because each action is a single command at a specific time.